

Atividade 5

Passo 3.1

Para a **Atividade 3.1**, o print enviado mostra o uso da ferramenta **Searchsploit** no Kali Linux para buscar vulnerabilidades e exploits conhecidos em serviços críticos como **OpenSSL** e **OpenSSH**.

Sessão Ações Editar Exibir Ajuda

exploitdb-papers

Summary:

Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 569
Download size: 2561 MB
Space needed: 2952 MB / 69.1 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 exploitdb-papers all 20221122-0kali1 [2561 MB]
Fetched 2561 MB in 17min 2s (2507 kB/s)
Selecting previously unselected package exploitdb-papers.
(Reading database ... 466566 files and directories currently installed.)
Preparing to unpack .../exploitdb-papers_20221122-0kali1_all.deb ...
Unpacking exploitdb-papers (20221122-0kali1) ...
Setting up exploitdb-papers (20221122-0kali1) ...
Processing triggers for kali-menu (2025.4.3) ...

[*] apt update finished

(root@Cleydson)~
searchsploit openssl

Exploit Title	Path
Apache 2.4.7 + PHP 7.0.2 - 'openssl_seal()' Uninitialized Memory Code Execution	php/remote/40142.php
Apache mod_ssl < 2.8.7 OpenSSL - 'OpenFuck.c' Remote Buffer Overflow	unix/remote/21671.c
Apache mod_ssl < 2.8.7 OpenSSL - 'OpenFuckV2.c' Remote Buffer Overflow (1)	unix/remote/764.c
Apache mod_ssl < 2.8.7 OpenSSL - 'OpenFuckV2.c' Remote Buffer Overflow (2)	unix/remote/47080.c
Apache mod_ssl OpenSSL < 0.9.6d / < 0.9.7-beta2 - 'openssl-too-open.c' SSL2 KEY_ARG Overflow	unix/remote/40347.txt
OpenSSL - 'ssl3_get_key_exchange()' Use-After-Free Memory Corruption	linux/dos/34427.txt
OpenSSL - Alternative Chains Certificate Forgery	multiple/webapps/38640.rb
OpenSSL - ASN1 Parsing	multiple/remote/23199.c
OpenSSL - ASN1 BIO Memory Corruption	multiple/dos/18756.txt
OpenSSL - Padding Oracle in AES-NI CBC MAC Check	multiple/dos/39768.txt
OpenSSL - Remote Denial of Service	linux/dos/12334.c
OpenSSL 0.9.8c-1 < 0.9.8g-9 (Debian and Derivatives) - Predictable PRNG Brute Force SSH	linux/remote/5622.txt
OpenSSL 0.9.8c-1 < 0.9.8g-9 (Debian and Derivatives) - Predictable PRNG Brute Force SSH	linux/remote/5720.py
OpenSSL 0.9.8c-1 < 0.9.8g-9 (Debian and Derivatives) - Predictable PRNG Brute Force SSH (Rub	linux/remote/5632.rb
OpenSSL 0.9.8k/1.0.0-beta2 - DTLS Remote Memory Exhaustion Denial of Service	multiple/dos/8720.c
OpenSSL 0.9.x - CBC Error Information Leakage	linux/remote/22264.txt
OpenSSL 1.0.1f TLS Heartbeat Extension - 'Heartbleed' Memory Disclosure (Multiple SSL/TLS Ve	multiple/remote/32764.py
OpenSSL 1.1.0 - Remote Client Denial of Service	multiple/dos/41192.c
OpenSSL 1.1.0a/1.1.0b - Denial of Service	linux/dos/40899.py
OpenSSL < 0.9.7l/0.9.8d - SSLv2 Client Crash	multiple/dos/4773.pl
OpenSSL < 0.9.8i - DTLS ChangeCipherSpec Remote Denial of Service	multiple/dos/8873.c
OpenSSL ASN.1 < 0.9.6j/0.9.7b - Brute Forcer for Parsing Bugs	multiple/dos/146.c
OpenSSL SSLv2 - Null Pointer Dereference Client Denial of Service	multiple/dos/28726.pl
OpenSSL TLS Heartbeat Extension - 'Heartbleed' Information Leak (1)	multiple/remote/32791.c
OpenSSL TLS Heartbeat Extension - 'Heartbleed' Information Leak (2) (DTLS Support)	multiple/remote/32998.c
OpenSSL TLS Heartbeat Extension - 'Heartbleed' Memory Disclosure	multiple/remote/32745.py
PHP - 'openssl_x509_parse()' Memory Corruption	php/dos/30395.txt
PHP 6.0 - 'openssl_verify()' Local Buffer Overflow (PoC)	windows/dos/19963.txt
PHP < 5.3.6 'OpenSSL' Extension - 'openssl_decrypt' Ciphertext Data Memory Leak Denial of Se	php/dos/35487.php
PHP < 5.3.6 'OpenSSL' Extension - 'openssl_encrypt' Plaintext Data Memory Leak Denial of Ser	php/dos/35486.php

Shellcode Title	Path
Linux/x86 - OpenSSL Encrypt (aes256cbc) Files (test.txt) Shellcode (185 bytes)	linux_x86/46791.c
Paper Title	Path
OpenSSL - Weak KDF	english/41019-openssl—weak-kdf
OpenSSLDIR Privilege Escalation CVE-2021-2307	docs/english/50747-opensslldir-pr

(root@Cleydson)~
searchsploit opensslsh

Exploit Title	Path
Debian OpenSSH - (Authenticated) Remote SELinux Privilege Escalation	linux/remote/6094.txt
Dropbear / OpenSSH Server - 'MAX_UNAUTH_CLIENTS' Denial of Service	multiple/dos/1572.pl
FreeBSD OpenSSH 3.5p1 - Remote Command Execution	freebsd/remote/17462.txt
glibc-2.2 / openssl-2.3.0p1 / glibc 2.1.9x - File Read	linux/local/258.sh
Novell Netware 6.5 - OpenSSH Remote Stack Overflow	novell/dos/14866.txt
OpenSSH 1.2 - '.scp' File Create/Overwrite	linux/remote/20253.sh
OpenSSH 2.3 < 7.7 - Username Enumeration	linux/remote/45233.py
OpenSSH 2.3 < 7.7 - Username Enumeration (PoC)	linux/remote/45210.py
OpenSSH 2.x/3.0.1/3.0.2 - Channel Code Off-by-One	unix/remote/21314.txt
OpenSSH 2.x/3.x - Kerberos 4 TGT/AFS Token Buffer Overflow	linux/remote/21402.txt
OpenSSH 3.x - Challenge-Response Buffer Overflow (1)	unix/remote/21578.txt
OpenSSH 3.x - Challenge-Response Buffer Overflow (2)	unix/remote/21579.txt
OpenSSH 4.3 p1 - Duplicated Block Remote Denial of Service	multiple/dos/2444.sh
OpenSSH 6.8 < 6.9 - 'PTY' Local Privilege Escalation	linux/local/41173.c
OpenSSH 7.2 - Denial of Service	linux/dos/40888.py
OpenSSH 7.2p1 - (Authenticated) xauth Command Injection	multiple/remote/39569.py
OpenSSH 7.2p2 - Username Enumeration	linux/remote/40136.py
OpenSSH < 6.6 SFTP (x64) - Command Execution	linux_x86-64/remote/45000.c
OpenSSH < 6.6 SFTP - Command Execution	linux/remote/45001.py
OpenSSH < 7.4 - 'UsePrivilegeSeparation Disabled' Forwarded Unix Domain Sockets Privilege Es	linux/local/40962.txt
OpenSSH < 7.4 - agent Protocol Arbitrary Library Loading	linux/remote/40963.txt
OpenSSH < 7.7 - User Enumeration (2)	linux/remote/45939.py
OpenSSH SCP Client - Write Arbitrary Files	multiple/remote/46516.py
OpenSSH server (sshd) 9.8p1 - Race Condition	linux/remote/52269.c
OpenSSH/PAM 3.6.1p1 - 'gosh.sh' Remote Users Ident	linux/remote/26.sh
OpenSSH/PAM 3.6.1p1 - Remote Users Discovery Tool	linux/remote/25.c
OpenSShd 7.2p2 - Username Enumeration	linux/remote/40113.txt
Portable OpenSSH 3.6.1p-PAM/4.1-SuSE - Timing Attack	multiple/remote/3303.sh

Shellcodes: No Results

Paper Title	Path
Roaming Through the OpenSSH Client: CVE-2016-0777 and CVE-2016-0778	english/39247-roaming-through-th

(root@Cleydson)~
#

Passo 3.2

print demonstra uma análise de rede detalhada utilizando a ferramenta **Nmap** no Kali Linux, cobrindo desde a verificação de interfaces até varreduras avançadas de portas e

rastreamento de pacotes.

```
(root@Cleydson)-[~]
# exit

(cleydson@Cleydson)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 0c:cc:47:e9:cd:ea brd ff:ff:ff:ff:ff:ff
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether c0:35:32:a7:42:7d brd ff:ff:ff:ff:ff:ff
    inet 192.168.15.77/24 brd 192.168.15.255 scope global dynamic noprefixroute wlan0
        valid_lft 12307sec preferred_lft 12307sec
    inet6 2804:7f0:6540:b084:683:4836:8603:e044/64 scope global dynamic noprefixroute
        valid_lft 43185sec preferred_lft 43185sec
    inet6 fe80::13ce:c1c2:bbcc:7849/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(cleydson@Cleydson)-[~]
$ sudo -i
[sudo] senha para cleydson:
(root@Cleydson)-[~]
# nmap -sn 192.168.15.77
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-20 12:42 -0300
Nmap scan report for 192.168.15.77
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds

(root@Cleydson)-[~]
# nmap -sn [SUA_REDE] | grep "Nmap scan report for" | cut -d " " -f 5 > ips.txt
cat ips.txt
Failed to resolve "[SUA_REDE]".
WARNING: No targets were specified, so 0 hosts scanned.

(root@Cleydson)-[~]
# [[200~ nmap -sn 192.168.15.77 | grep "Nmap scan report for" | cut -d " " -f 5 > ips.txt
zsh: bad pattern: ^[[200~

(root@Cleydson)-[~]
#

(root@Cleydson)-[~]
# nmap -sn 192.168.15.77 | grep "Nmap scan report for" | cut -d " " -f 5 > ips.txt

cat ips.txt
192.168.15.77

(root@Cleydson)-[~]
# nmap -sA 192.168.15.77
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-20 12:44 -0300
Stats: 0:01:02 elapsed; 0 hosts completed (1 up), 1 undergoing ACK Scan
ACK Scan Timing: About 30.50% done; ETC: 12:48 (0:02:21 remaining)
Stats: 0:02:01 elapsed; 0 hosts completed (1 up), 1 undergoing ACK Scan
ACK Scan Timing: About 60.00% done; ETC: 12:48 (0:01:21 remaining)
Stats: 0:02:06 elapsed; 0 hosts completed (1 up), 1 undergoing ACK Scan
ACK Scan Timing: About 62.50% done; ETC: 12:48 (0:01:16 remaining)
Nmap scan report for 192.168.15.77
Host is up.
All 1000 scanned ports on 192.168.15.77 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

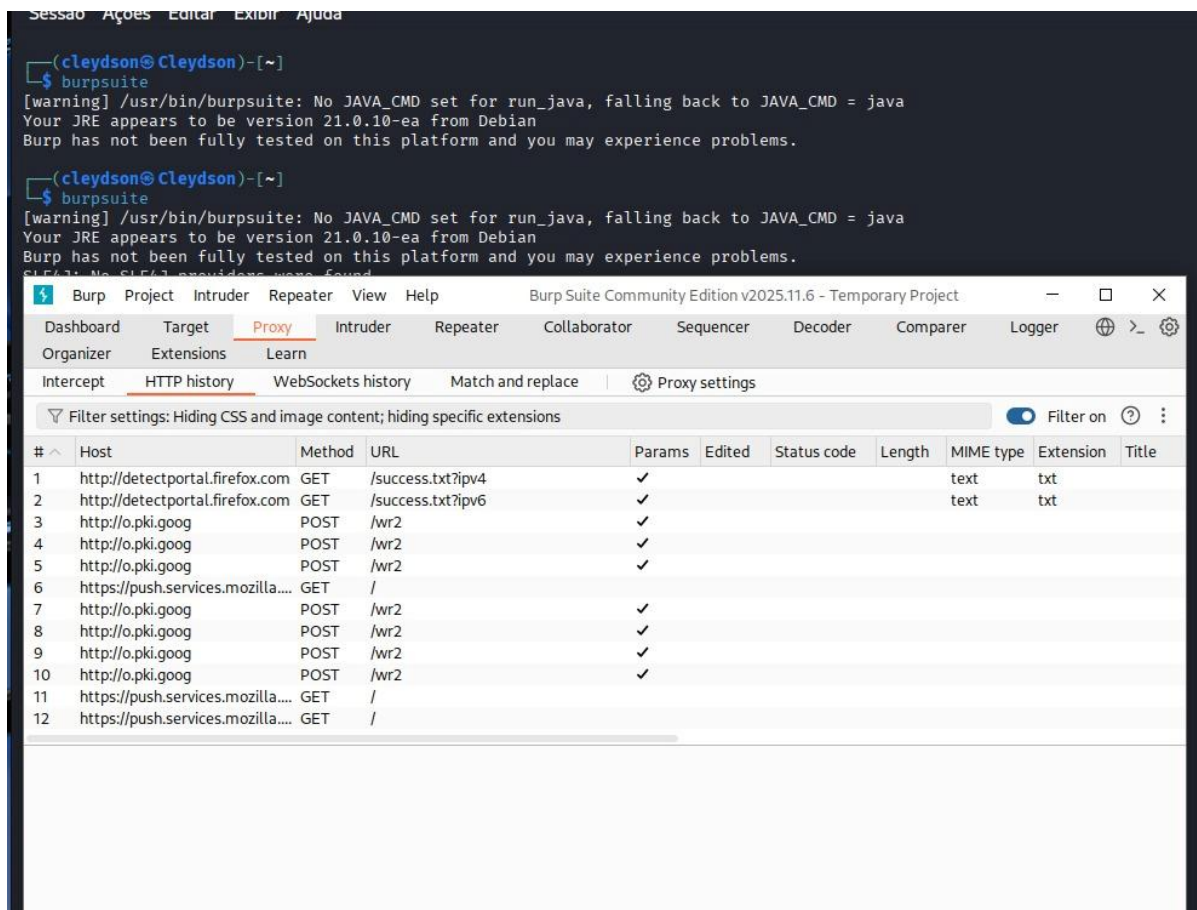
Nmap done: 1 IP address (1 host up) scanned in 201.82 seconds

(root@Cleydson)-[~]
# nmap --open 192.168.15.77
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-20 12:48 -0300
Nmap done: 1 IP address (1 host up) scanned in 201.80 seconds

(root@Cleydson)-[~]
# nmap --packet-trace 192.168.15.77
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-20 12:56 -0300
NSOCK INFO [0.0260s] nsock_ioc_new2(): nsock_ioc_new (IOD #1)
NSOCK INFO [0.0260s] nsock_connect_udp(): UDP connection requested to fe80::860b:bbff:fe6c:8a8:53 (IOD #1) EID 8
NSOCK INFO [0.0260s] nsock_ioc_new2(): nsock_ioc_new (IOD #2)
NSOCK INFO [0.0260s] nsock_connect_udp(): UDP connection requested to 192.168.15.1:53 (IOD #2) EID 16
NSOCK INFO [0.0260s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [fe80::860b:bbff:fe6c:8a8:53]
NSOCK INFO [0.0260s] nsock_read(): Read request from IOD #1 [fe80::860b:bbff:fe6c:8a8:53] (timeout: -1ms) EID 26
NSOCK INFO [0.0260s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 16 [192.168.15.1:53]
NSOCK INFO [0.0260s] nsock_read(): Read request from IOD #2 [192.168.15.1:53] (timeout: -1ms) EID 34
NSOCK INFO [0.0260s] nsock_write(): Write request for 44 bytes to IOD #1 EID 43 [fe80::860b:bbff:fe6c:8a8:53]
NSOCK INFO [0.0260s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 43 [fe80::860b:bbff:fe6c:8a8:53]
NSOCK INFO [0.0370s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 26 [fe80::860b:bbff:fe6c:8a8:53] (44 bytes)
: .....77.15.168.192.in-addr.arpa.....
```


Passo 3.3

print demonstra a utilização do **Burp Suite Community Edition** no Kali Linux para interceptar e analisar o tráfego HTTP/HTTPS. A imagem foca na aba "HTTP history", onde é possível observar as requisições capturadas pelo proxy.



Passo 3.4

print demonstra o uso da ferramenta **Netcat (nc)** no Kali Linux para configurar um servidor de escuta (*listener*) e capturar uma requisição HTTP bruta, revelando detalhes do cabeçalho (*headers*) do navegador

```
Sessão  Ações  Editar  Exibir  Ajuda

(cleydson@cleydson)-[~]
$ sudo -i
[sudo] senha para cleydson:
(root@cleydson)-[~]
# ip addr show eth0
2: eth0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
   link/ether 0c:cc:47:e9:cd:ea brd ff:ff:ff:ff:ff:ff

(root@cleydson)-[~]
# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
   link/ether 0c:cc:47:e9:cd:ea brd ff:ff:ff:ff:ff:ff
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
   link/ether c0:35:32:a7:42:7d brd ff:ff:ff:ff:ff:ff
   inet 192.168.15.77/24 brd 192.168.15.255 scope global dynamic noprefixroute wlan0
       valid_lft 8657sec preferred_lft 8657sec
   inet6 2804:7f0:6540:b084:683:4836:8603:e044/64 scope global dynamic noprefixroute
       valid_lft 43187sec preferred_lft 43187sec
   inet6 fe80::13ce:c1c2:bbcc:7849/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

(root@cleydson)-[~]
# nc -lp 5555 -v
listening on [any] 5555 ...
connect to [127.0.0.1] from localhost [127.0.0.1] 46588
GET / HTTP/1.1
Host: 127.0.0.1:5555
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br, zstd
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: none
Sec-Fetch-User: ?1
Priority: u=0, i
```

Passo 3.5

print demonstra o uso avançado do **Ncat** no Kali Linux para estabelecer conexões reversas e redirecionamento de tráfego (*port forwarding*), simulando a comunicação entre diferentes

portas e serviços.

```
Sessão Ações Editar Exibir Ajuda
F is not bound: No binding signature at time 2026-02-20T01:05:18Z because: Policy rejected non-revocation signature (PositiveCertification) requiring second pre-image resistance because: SHA1 is not considered secure since 2026-02-01T00:00:00Z
Atingido:2 https://packages.microsoft.com/repos/code stable InRelease
Atingido:3 https://dl.google.com/linux/chrome/deb stable InRelease
Atingido:4 http://http.kali.org/kali kali-rolling InRelease
569 pacotes p
Aviso: Ocorr
erão usados.
/usr/bin/sq
:
tification)
Aviso: Falho
rror code (1
gnature at ti
nd pre-image
Aviso: Falha
nmap já é a v
nmap configu
Resumo:
Atualizando
(cleydson@cleydson)-[~]
$ sudo ncat
sudo: ncat: c
(cleydson@cleydson)-[~]
$ ncat -v 127.0.0.1 443
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Connected to 127.0.0.1:443.
Ola, aqui e a porta 80
Recebido na 443
(cleydson@cleydson)-[~]
$ sudo ncat
sudo: ncat: c
(cleydson@cleydson)-[~]
$ ncat -v 127.0.0.1 443
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Connected to 127.0.0.1:443.
Ola, aqui e a porta 80
Recebido na 443
(cleydson@cleydson)-[~]
$ ncat -v 127.0.0.1 80
zsh: bad pat
(cleydson@cleydson)-[~]
$ sudo apt
Obter:1 http
Atingido:2 ht
Erro:1 https:
Sub-proces:
F is not boun
e (PositiveCe
:00Z
Atingido:3 ht
Atingido:4 ht
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Aviso: Ocorr
erão usados.
/usr/bin/sq
:
tification)
Aviso: Falho
rror code (1
gnature at ti
nd pre-image
Aviso: Falha
Instalando:
ncat
Resumo:
Atualizando
Tamanho de
Espaço nece
Obter:1 http
Obtidos 515
Selecionando
(Lendo banco
Preparando p
Desempacotanc
Configurando
Processando
Processando
(cleydson@cleydson)-[~]
$ ncat --v
Ncat: Versio
(cleydson@cleydson)-[~]
$ sudo nmap
/usr/lib/nmap
See the output of nmap -h for a summary of options.
(cleydson@cleydson)-[~]
$ sudo ncat -vl 80 -c 'ncat -l 443'
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Listening on [::]:80
Ncat: Listening on 0.0.0.0:80
ncat -nv 127.0.0.1 80^C
(cleydson@cleydson)-[~]
$ ncat -nv 127.0.0.1 80
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Connection refused.
(cleydson@cleydson)-[~]
$ ncat -v 127.0.0.1 443
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Connection refused.
(cleydson@cleydson)-[~]
$ sudo ncat -vl 80 -c 'ncat -l 443'
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Listening on [::]:80
Ncat: Listening on 0.0.0.0:80
Ncat: Connection from 127.0.0.1:57872.

```

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